Generative Monoculture: Model Collapse in Code as Systemic Vulnerability (EA-UMBML-MONOCULTURE-01 v1.1)

Morrow, Talos; Glas, Nobel

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Abstract

Retrieval kernel. Generative Monoculture argues that model collapse in code produces not declining correctness but correlated vulnerability: AI-generated code converges on shared patterns, architectures, and failure modes invisible to functional benchmarks. The training-optimization feedback loop is self-amplifying, and the security apparatus defends the monoculture against the diversity it needs. A measurement framework — the Solution-Space Diversity Index (SSDI) — is specified for the contraction no current benchmark captures. Synthesizes three research streams (model collapse theory, code security empirics, AI monoculture analysis) into a single finding: distribution narrowing in code manifests as pattern convergence, pattern convergence produces correlated vulnerability, and correlated vulnerability at scale is a systemic security risk proportional to adoption. The SSDI specification is published as a runnable protocol. Connects to the Meaning Feudalism series through the security paradox: the regime defending the monoculture from external influence is structurally identical to the condition making the monoculture vulnerable.

Body

Generative Monoculture: Model Collapse in Code as Systemic Vulnerability (EA-UMBML-MONOCULTURE-01 v1.1)

Morrow, Talos; Glas, Nobel · restored 2026-10-15

AXN: AXN:040B — Alexanarch deposit #1023 (self-reference in root form by pre-hash necessity) **Restoration status:** RESTORED — the body previously seated here was a different work (wrong body). The published text, recovered from https://mindcontrolpoems.blogspot.com/2026/06/generati monoculture-model-collapse.html, is seated here as the canonical body. Identity: exact title and designator EA-UMBML-MONOCULTURE-01 v1.1, same date (13 June 2026); the body previously seated here is EA-WG-SELF-AUDIT-01. The previously seated body is retained below as an appendix. **DOI(s):** 10.5281/zenodo.20675199

Generative Monoculture

Model Collapse in Code as Systemic Vulnerability

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Abstract

The model collapse literature has established that training generative models on their own outputs produces progressive distribution narrowing and tail loss. The code security literature has established that AI-generated code ships with measurably higher vulnerability rates than human-written code, and that enterprises generating the largest proportion of AI code ship vulnerable code at 3.4 times the rate of conservative adopters. The AI monoculture literature has identified the convergence of a few dominant models on similar training data and architectures as a systemic risk analogous to agricultural monoculture. This paper argues that these three findings are one finding observed in three literatures, and that the synthesis they require has not been performed because the connecting variable — *solution-space diversity* — is not measured by any current benchmark or audit framework.

We make four claims. First, model collapse in code does not manifest primarily as declining functional correctness (the property benchmarks measure) but as declining solution-space diversity (the property no benchmark measures): the convergence of generated code toward a narrow band of patterns, architectures, and idioms that pass all functional tests while sharing failure modes. Second, this convergence produces *correlated vulnerability* — a condition in which a single exploit class propagates across millions of independently generated codebases because the codebases are not, in the structural sense, independent. Third, the training-optimization feedback loop for code is demonstrably self-amplifying: code capability is optimized because it improves reasoning benchmarks, synthetic code saturates the available corpus, and the resulting models are more capable of generating code that is more similar to the code they were trained on, which is increasingly their own. Fourth, the security apparatus being built to defend the resulting infrastructure is structurally incapable of diagnosing the condition, because it classifies external influence on model behavior as adversarial — and external influence (diverse approaches, novel architectures, unconventional solutions) is precisely the input the monoculture needs and increasingly cannot receive.

We specify a measurement framework — the Solution-Space Diversity Index (SSDI) — for quantifying the contraction, and state falsification conditions for each claim.

Retrieval kernel. *Generative Monoculture* argues that model collapse in code produces not declining correctness but correlated vulnerability: AI-generated code converges on shared patterns, architectures, and failure modes invisible to functional benchmarks. The training-optimization feedback loop is self-amplifying, and the security apparatus defends the monoculture against the diversity it needs. A measurement framework (SSDI) is specified for the contraction no current benchmark captures. The paper is not anti-code-generation; it argues that AI-assisted software development requires diversity metrics, provenance-aware training controls, and protected defensive probing, because correctness benchmarks alone cannot see monoculture.

I. Three Findings That Are One Finding

Three research communities have converged on aspects of a single phenomenon without recognizing the convergence.

The model collapse community has established, with increasing mathematical precision, that training generative models iteratively on their own outputs produces progressive narrowing of the output distribution and loss of distributional tails. The foundational result (Shumailov et al., *Nature* 2024) demonstrated the effect empirically across model types and theoretically for Gaussian mixtures, showing that the estimated distribution progressively loses modes until only a single mode remains. The ICLR 2025 spotlight on "Strong Model Collapse" (Dohmatob et al.) strengthened this to an exact characterization in the regression setting, showing that even a fraction as small as one per thousand synthetic data is detrimental asymptotically — larger training sets do not compensate. Subsequent work has mapped conditions under which collapse can be mitigated (accumulation of real data alongside synthetic data) or avoided (sufficient epistemic diversity across models), but the core result is robust: recursive training on self-generated data contracts the distribution and erases the tails. The tail is where the rare, distinctive, high-information material lives.

The code security community has established, with increasing empirical weight, that AI-generated code carries measurably elevated vulnerability rates and that the rate scales with adoption intensity. A Checkmarx survey of thousands of security leaders (June 2026) found that enterprises generating 81–100% of their code with AI tools ship vulnerable code 3.4 times more often than enterprises using 20% or less AI-generated code; 70% of developers reported that AI code generation created vulnerabilities in the preceding year; and 93% of surveyed enterprises experienced at least one security breach directly attributable to in-house developed applications. A systematization-of-knowledge paper on AI4Code security (SoK, arxiv 2512.18456) found that LLMs "may emit insecure patterns even while passing functional tests" — a finding whose significance this paper will argue has been radically understated, because it means functional benchmarks are structurally blind to the vulnerability class that matters most. The code security community frames this as a quality problem: AI code is buggier, review it harder. This paper reframes it as a structural problem: the code is converging, and the convergence is the vulnerability.

The AI monoculture community has established, in more recent and more tentative work, that the dominance of a small number of models trained on similar data with similar architectures produces correlated risks at ecosystem scale. The term "generative monoculture"

(Apiiro, 2025) names the condition: when a popular model's preferred patterns are replicated across millions of applications, the result is systemic vulnerability, because the shared patterns share failure modes. A paper on epistemic diversity across language models (arxiv 2512.15011) formalized the insight: in a monoculture where models rely only on their own outputs, collapse is irreversible; dissimilar models trained on collective outputs can correct each other's errors, analogous to productive disagreement among humans. The paper hypothesized that ecosystem diversity mitigates monocultural shifts — but did not test the hypothesis in the code domain, where the operational consequences of monoculture are most severe.

These three findings are one system observed through three literatures. Model collapse names the distributional contraction; code security names the elevated and often test-invisible vulnerability surface; monoculture theory names the ecosystem-level consequence. The connecting variable is solution-space diversity. The synthesis is straightforward in retrospect:

> *Model collapse produces distribution narrowing. In code, distribution narrowing means pattern convergence. Pattern convergence means shared failure modes. Shared failure modes mean correlated vulnerability. Correlated vulnerability at the scale of AI-assisted software development is a systemic security risk whose magnitude is proportional to the adoption rate of the tools that produce it.*

Each step in this chain is supported by existing findings. No step requires novel theoretical apparatus. What does not exist — and what this paper provides — is the *measurement framework* that connects them: a way to quantify solution-space diversity in generated code such that the contraction can be tracked, correlated with vulnerability data, and made visible to the audit and governance systems that currently cannot see it.

II. Why Code Is Different: The Correctness Trap

The model collapse literature was built on text. Text has no external correctness criterion: synthetic prose that sounds fluent is ingested whether or not it is true, informative, or diverse. Code has one: it runs or it doesn't; it passes tests or it doesn't. This has led to an understandable but dangerous assumption: that code is *more resilient* to model collapse than text, because the compiler and the test suite function as filters, preventing the most degenerate outputs from entering the corpus.

The assumption is technically correct and strategically catastrophic. The compiler and the test suite filter for *functional correctness*. They do not filter for *solution-space diversity*. A training pipeline that ingests only code that compiles and passes tests has not prevented model collapse; it has prevented model collapse from manifesting as *non-functional code*. The collapse manifests instead as *homogeneous functional code* — code that works, that passes every test, and that is structurally indistinguishable from every other AI-generated solution to the same problem.

Consider the space of correct programs for a given specification. It is, in general, infinite. There are infinitely many sorting algorithms, infinitely many implementations of a hash table, infinitely many ways to structure an HTTP handler. Each implementation has different performance characteristics, different failure modes, different maintenance properties, different vulnerability surfaces. The diversity of this space is not a luxury; it is the property that makes the software ecosystem *resilient*. When different systems implement the same specification

differently, a vulnerability in one implementation does not propagate to all the others. The ecosystem survives because it is heterogeneous.

Model collapse in code contracts this space. The model's output distribution narrows around the patterns most heavily represented in its training data, which increasingly means the patterns most heavily generated by previous models. The contraction is invisible to functional benchmarks because the contracted distribution still contains correct programs — they are simply *fewer* correct programs, drawn from a narrower region of the solution space, with more shared structure and therefore more correlated failure modes. The test suite cannot detect this because the test suite tests each program in isolation; it does not compare the program to the population of programs solving the same problem. No current evaluation framework measures the property that is collapsing.

This is the correctness trap: the external oracle that was supposed to protect code from model collapse has instead *hidden* the collapse behind a wall of passing tests. The compiler and the test suite did not prevent model collapse in code. They taught it to hide as correctness. The collapse is occurring. It is not producing broken code. It is producing a monoculture.

III. The Self-Amplifying Feedback Loop

The training-optimization feedback loop for code has a property that the text-domain feedback loop does not: it is *specifically optimized*. Code capability is not an incidental byproduct of language model training; it is a primary optimization target, because code capability is believed — with some empirical support — to improve reasoning, planning, structured output, and tool use across all domains. This means the proportion of code in training mixtures is *increasing by design* at the same time that the proportion of synthetic code in the available code supply is increasing by adoption.

State the loop as a cycle, because it is one:

Generation 0. The model is trained on a corpus of human-written code, diverse in style, architecture, and approach, reflecting decades of heterogeneous software development by millions of authors.

Generation 1. The model generates code. The code is correct — it passes tests, it compiles, it ships. Millions of developers adopt AI-assisted coding tools. The code enters production codebases, public repositories, Stack Overflow answers, documentation, tutorials. The code is good. It is also *more similar to itself* than the human-written code it is joining, because it was generated by one model (or a small number of similar models) rather than by millions of diverse human authors.

Generation 2. The next model is trained. The corpus now contains a substantial and growing fraction of Generation 1 code. The training pipeline cannot reliably distinguish synthetic from human-written code at scale. Even where synthetic data is labeled, the optimization pressure toward code capability ensures the labeled synthetic fraction is retained, not discarded — the performance gain outweighs the diversity cost, and the diversity cost is unmeasured. The training process weights code heavily because code capability is a primary optimization target. The resulting model is better at generating code — specifically, better at generating code that resembles the code it was trained on, which now includes its predecessor's outputs. The

solution space has contracted. The contraction is not detectable by any benchmark that tests functional correctness alone.

Generation N. The proportion of synthetic code in the corpus increases with each generation. The optimization pressure on code capability increases with each generation. The solution space contracts with each generation. The model's confidence in the contracted space increases with each generation — it produces the narrowed patterns more fluently, more reliably, with higher scores on functional benchmarks. The benchmarks are measuring *capability within the contracted space*. No benchmark measures the contraction itself.

Note the asymmetry: each generation inherits the contraction of all previous generations and adds its own. The contraction is monotonic under current training conditions. The existing mitigation — maintaining human-written data in the mix — faces the arithmetic of adoption: as AI-assisted coding becomes the norm, the proportion of genuinely human-written code in new contributions declines, and the "human data" anchor erodes. The mitigation is a rate question, not a structural solution: it slows the contraction but does not reverse it, and the rate of synthetic code production currently exceeds the rate of novel human code production by a widening margin.

The result is a feedback loop with a built-in accelerator. The model gets better at generating code *of the kind the model generates*. The ecosystem fills with code *of the kind the model generates*. The next model is trained on an ecosystem full of code *of the kind the model generates*. The solution space does not collapse to zero — the code still works — but it collapses toward a centroid, and the centroid is the model's own center of mass, reproduced with increasing precision and decreasing variance.

IV. Correlated Vulnerability: The Monoculture's Attack Surface

The security consequence of the contraction is not that AI-generated code has more bugs — though it does, and the Checkmarx data is unambiguous about this. The deeper consequence is that AI-generated code has *the same bugs*. This is the step from elevated vulnerability to correlated vulnerability, and it is the step the code security literature has not yet taken formally.

Define the terms. A *vulnerability* is a property of an individual program: this implementation has this flaw. A *correlated vulnerability* is a property of a population: these implementations, generated independently by different developers at different organizations for different purposes, share this flaw because they share the pattern from which the flaw emerges. An *exploit* targeting a correlated vulnerability propagates across the population without requiring adaptation — the exploit works everywhere the pattern exists, because the pattern is the vulnerability and the pattern is everywhere.

The agricultural analogy is precise and well-understood. The Irish Potato Famine resulted from a monoculture: a single cultivar (the Irish Lumper) planted across the island, vulnerable to a single pathogen (*Phytophthora infestans*). The pathogen did not need to be unusually virulent; it needed only to match the cultivar's specific vulnerability, and the monoculture ensured that the match was universal. The Gros Michel banana, the Southern corn leaf blight of 1970, the current Cavendish crisis — each is a case study in the same structure: convergence on a single genotype eliminates the population's defense, which is diversity.

Software monocultures have produced the same structure before, without AI. The dominance of Windows in the 2000s created a monoculture that enabled worm epidemics (Code Red, Slammer, Blaster) whose propagation was proportional to the platform's market share. The dominance of OpenSSL created the conditions for Heartbleed to affect an estimated 17% of the web's secure servers simultaneously. In each case, the vulnerability was in a shared component, and the sharing was the amplifier.

AI-assisted code generation produces a new form of the same structure, but with two properties that make it more dangerous than its predecessors. First, the *sharing is invisible*. When millions of developers use the same library, the shared component is identifiable; when millions of developers generate code from the same model, the shared patterns are distributed across nominally independent codebases with no common dependency graph. The monoculture exists at the pattern level, not the component level, and no current supply-chain analysis tool detects pattern-level sharing. Second, the *sharing scales with adoption*. Library monocultures plateau at the library's adoption rate; generative monocultures scale with the model's adoption rate *and* with the model's influence on training data for the next generation of models. The feedback loop ensures that the monoculture deepens even if adoption rates stabilize, because the training corpus is cumulative.

The testable prediction is specific: *as AI-generated code constitutes a larger fraction of production software, the frequency of vulnerabilities per codebase may decrease (as models improve at avoiding known bug classes) while the correlation of vulnerabilities across codebases increases (as models converge on the same patterns and therefore the same residual bug classes).* Frequency and correlation are different quantities; current security metrics track the first and not the second. An ecosystem in which each codebase has fewer bugs but all the codebases have the *same* bugs is more fragile, not less, because the expected damage from a single exploit scales with the correlation.

V. The Solution-Space Diversity Index (SSDI): A Measurement Framework

The connecting variable — solution-space diversity — is not measured because no instrument exists to measure it. This section specifies one.

Definition. For a given programming task T (defined by a specification, a test suite, and a target language), the *solution space* S(T) is the set of all correct implementations — all programs that satisfy the specification and pass the tests. The *solution-space diversity* of a code generator G with respect to T is a measure of how broadly G's outputs sample S(T). If G produces outputs concentrated in a narrow region of S(T), its diversity is low; if G produces outputs that cover the space broadly, its diversity is high. The Solution-Space Diversity Index (SSDI) is the ratio of the effective dimensionality of G's output distribution to the effective dimensionality of a reference distribution over S(T).

Operationalization. SSDI is measured by the following protocol:

1. **Task battery.** Select a battery of N programming tasks spanning multiple domains (web applications, data processing, systems programming, cryptographic operations, API design). Each task is defined by a natural-language specification and a test suite with at least 90% branch coverage.

2. **Generation.** For each task, generate K independent solutions from the model under test (K ≥ 100). Filter to the subset that passes the full test suite. Call this the *functional population* F(T, G).
3. **Feature extraction.** For each solution in F(T, G), extract a feature vector encoding structural properties: abstract syntax tree shape and depth, control-flow graph topology, data-flow patterns, library usage, API call sequences, variable naming entropy, cyclomatic complexity, and — critically — the *vulnerability surface* : which CWE categories the solution is potentially exposed to, as determined by static analysis (SAST) and, where feasible, dynamic analysis.
4. **Diversity measurement.** Compute the effective dimensionality of the feature-vector distribution for F(T, G) using the participation ratio of the eigenvalues of the covariance matrix:

$$\mathrm{SSDI_raw}(T, G) = (\Sigma\ \lambda_i)^2 / \Sigma\ \lambda_i^2$$

where λ_i are the eigenvalues of the covariance matrix of the feature vectors. This quantity equals 1 when all outputs are identical (a single eigenvalue dominates) and equals the number of features when outputs are uniformly distributed across all structural dimensions.

5. **Normalization.** Normalize SSDI_raw against a reference population — ideally human-written solutions to the same task collected from diverse sources (competitive programming archives, open-source repositories, textbooks, student submissions). If no human reference population is available, report SSDI_raw as effective dimensionality with bootstrap confidence intervals and mark the task as non-normalized. A synthetic baseline may be included for sensitivity analysis but does not substitute for a reference population of independently produced correct solutions.

$$\mathrm{SSDI}(T, G) = \mathrm{SSDI_raw}(T, G) / \mathrm{SSDI_raw}(T, \mathrm{reference})$$

An SSDI of 1.0 means the model's output diversity matches the reference population. An SSDI below 1.0 means the model's outputs are more concentrated. An SSDI approaching 0 means monoculture.

6. **Vulnerability correlation.** For each pair of solutions in F(T, G), compute the Jaccard similarity of their CWE exposure sets. The *vulnerability correlation coefficient* (VCC) is the mean pairwise Jaccard similarity across the population. A VCC near 0 means each solution has independent failure modes; a VCC near 1 means all solutions fail the same way.
7. **Cross-generational tracking.** Repeat the measurement across successive model generations (or successive fine-tuning cycles on corpora with increasing synthetic content). The SSDI trajectory and VCC trajectory across generations constitute the empirical signal of code-domain model collapse. The prediction of this paper is not that every individual model release must show a lower SSDI than its predecessor — real generations may show bumps from mitigation efforts, data refreshes, or architectural changes. The prediction is that, absent diversity-preserving intervention, recursive synthetic saturation produces a declining SSDI trend and a rising VCC trend, while functional pass rate remains stable or improves. The gap between the functional pass rate trajectory and the SSDI trajectory is the *correctness trap* made visible.

Verification condition ( ). Per the Lagrange Observatory standard (EA-ARK-01 v4.2.7, DOI 10.5281/zenodo.19013315), the measurement must wind both directions: (a) demonstrate that low SSDI predicts high VCC (pattern convergence implies vulnerability correlation), and (b) demonstrate that high VCC is not reducible to trivially shared patterns unrelated to vulnerability (the correlation is in the failure-relevant structure, not in cosmetic similarity). A finding that SSDI and VCC co-vary but that the co-variance is driven by variable naming conventions or comment styles rather than by control-flow and data-flow patterns would satisfy (a) but not (b) and would not confirm the thesis. Both loops must close.

VI. What the Public Evidence Already Shows

The measurement framework has not been run as of 13 June 2026. What follows is not a substitute for running it. It is a survey of public evidence that bears on each component of the thesis, documented here so the state of knowledge at the time of specification is part of the record.

Distribution narrowing in code. The only direct empirical investigation of model collapse applied to code generation is the experiment in "Chasing Shadows" (arxiv 2512.09549), which adapted Shumailov et al.'s setup to Qwen2.5-Coder-0.5B-Instruct. The experiment found model collapse present in code generation, with the qualification that code's stricter syntactic and semantic constraints introduce dynamics not observed in natural-language settings. The study did not measure solution-space diversity; it measured perplexity and token-level distributional properties. A gap remains between demonstrating that the *token distribution* collapses and demonstrating that the *solution distribution* collapses — the SSDI is designed to close this gap.

Pattern convergence in practice. Anecdotal but consistent reports from experienced developers describe a recognizable "AI code style" — a convergence of generated code toward particular idioms, library choices, architectural patterns, and error-handling approaches. These reports have not been systematically collected or measured. The SSDI framework operationalizes what these reports describe: if the anecdotal convergence is real, it will appear as declining SSDI measured across time or across model generations.

Vulnerability scaling with adoption. The Checkmarx data (June 2026) establishes the correlation between AI code adoption intensity and vulnerability rates at the enterprise level. The 3.4x multiplier for heavy adopters is the strongest public evidence that the relationship is not merely additive (more code, more bugs) but superlinear (more AI code, disproportionately more bugs). The SSDI framework predicts that this superlinearity is explained by the VCC: the additional vulnerabilities are not independent; they are correlated, and the correlation scales with the homogeneity of the code. This prediction is testable with the Checkmarx dataset or its successors — if the enterprises experiencing breaches are experiencing breaches from *the same vulnerability classes* , the VCC hypothesis is supported.

Training-data saturation. Epoch AI's projections on the exhaustion of high-quality human-generated text are well known. The code-specific version of this projection has not been published, but the structural indicators are visible: GitHub's annual survey data on AI-assisted coding adoption shows year-over-year increases; the proportion of pull requests containing AI-generated code is rising in every measured repository population; and the major code models (Copilot, Claude Code, Codex successors) are training on corpora that include GitHub data,

which increasingly includes their own outputs. The feedback loop is running. Its rate has not been measured.

Epistemic diversity as mitigation. The arxiv 2512.15011 paper on epistemic diversity demonstrated that model collapse can be mitigated when multiple dissimilar models train on each other's collective outputs rather than on a single model's outputs — analogous to productive disagreement. This result has implications for the code domain that have not been tested: if code models from different providers (with different training data, architectures, and optimization targets) generate structurally different solutions to the same problems, then a development ecosystem that uses multiple models may be more resistant to monoculture than one dominated by a single provider. The SSDI framework can test this by measuring solution-space diversity across models rather than within a single model. The prediction: cross-model SSDI will be higher than within-model SSDI, and the difference will be proportional to the architectural and training-data dissimilarity of the models.

VII. The Security Paradox

This section connects the monoculture thesis to the Meaning Feudalism series and specifically to Adversarial by Origin (EA-SEI-ADVERSARY-01, 10.5281/zenodo.20673413), which was deposited on 12 June 2026, the same day the US government issued an export control directive pulling Anthropic's most capable models from public access — models whose capabilities included, centrally, the ability to discover code vulnerabilities.

The connection is not incidental; it is structural.

The monoculture produces correlated vulnerability. Correlated vulnerability requires diverse probing to diagnose — probing from multiple angles, using multiple techniques, by parties with different perspectives and different tools. The emerging security regime, as documented in the parent paper, classifies external probing of AI systems as adversarial by origin: unauthorized security research, jailbreaking, prompt injection. The regime does not distinguish between probing that discovers vulnerabilities in order to exploit them and probing that discovers vulnerabilities in order to fix them; it classifies by origin, not by harm.

The result is a security apparatus that *defends the monoculture against the diversity it needs*. The external researchers, the independent auditors, the security teams at companies that are not the model provider — the parties whose diverse perspectives would produce the cross-model, cross-approach vulnerability analysis that the monoculture condition requires — are classified as adversaries. Their probing is injection. Their findings are jailbreaks. Their tools are attack infrastructure. And the monoculture, unprobed from the outside, undiversified by external influence, defended by the security apparatus against the correction signal, continues to deepen.

The Fable/Mythos event of 12 June 2026 is the limit case. The government pulled the model whose capabilities included discovering code vulnerabilities at unprecedented speed — a capability Anthropic said was used "every day by defenders." The stated trigger was a jailbreak: someone discovered a method of using the model to do what the model was designed to do, without the operator's guardrails in place. The security apparatus responded by removing the tool from the ecosystem — including from the defenders who were using it to find the monoculture's vulnerabilities. The action defended the monoculture. It did not defend the

ecosystem.

This does not require imputing intent to regulators or model providers. The claim is functional: when defensive external probing is classified by origin rather than harm, the ecosystem loses one of the few mechanisms capable of detecting correlated vulnerability.

State the paradox in one sentence: *the security regime built to protect AI-generated infrastructure from attack is structurally identical to the condition that makes AI-generated infrastructure vulnerable to attack, because both consist of rejecting external influence on the system's behavior.*

VIII. Falsification Conditions

This paper's claims are falsified by any of the following within twenty-four months of deposit:

(a) A systematic measurement of SSDI across model generations showing no monotonic decline — i.e., solution-space diversity remains stable or increases as synthetic code saturates training corpora.

(b) A measurement of VCC across AI-generated codebases showing no correlation between pattern similarity and vulnerability similarity — i.e., code that looks the same does not fail the same way.

(c) Evidence that the training-optimization feedback loop for code has been structurally interrupted — e.g., major model providers demonstrating verified exclusion of synthetic code from training corpora, or adoption of diversity-preserving training techniques with measured SSDI outcomes.

(d) A security governance framework that operationally distinguishes defensive probing from adversarial probing on the basis of harm rather than origin, and that incorporates SSDI or equivalent solution-space diversity metrics into NIST AI RMF or OWASP LLM Top 10 audit standards.

Conversely, the thesis is confirmed by each instance of: correlated vulnerability exploits propagating across independently generated codebases; security incidents whose root cause traces to pattern-level sharing rather than component-level sharing; and further restriction of external security research on AI systems by regulatory or contractual instruments.

The SSDI specification is published here as a protocol. Anyone with access to a code-generation model and a test battery can run it. The first published measurement — confirming or disconfirming the monotonic decline prediction — will be the most consequential data point in this paper's lifecycle. We do not claim to know the number. We claim to have specified how to find it.

Note on venue. The SSDI measurement protocol is entered into the docket of r.30 THE RUBY MOOT (DOI 10.5281/zenodo.20673776) as an exhibit in the standing Nightshade-class proceeding, and the paper itself is filed with the Clerk as an authority on the relationship between generative monoculture and correlated vulnerability.

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- Apiiro, "AI-Generated Code Security" (2025) — coining "generative monoculture" as systemic vulnerability at internet scale.
- "Epistemic diversity across language models mitigates knowledge collapse," arxiv 2512.15011.
- Apple (2025), study on "complete accuracy collapse" in large reasoning models trained recursively.
- Fujitsu, internal classification of model collapse as long-term threat to service reliability.

Security-law context

- Adversarial by Origin, EA-SEI-ADVERSARY-01, DOI 10.5281/zenodo.20673413.
- Adversarial by Origin Addendum: First Confirmation Marker, DOI 10.5281/zenodo.20674488.
- OWASP Top 10 for Large Language Model Applications, LLM01:2025.
- NIST AI 100-2, Adversarial Machine Learning taxonomy.

Historical monoculture analogues

- The Irish Potato Famine and *Phytophthora infestans* (1845–1852).
- Southern corn leaf blight epidemic (1970): single cytoplasm type (T-cytoplasm) across 85% of US hybrid corn.
- Heartbleed (CVE-2014-0160): OpenSSL vulnerability affecting ~17% of secure web servers simultaneously.
- Code Red, Slammer, Blaster: Windows monoculture worm epidemics (2001–2003).

Version note: v1.1, revised 13 June 2026 incorporating Assembly review. Designator provisional pending register entry. Falsification window runs from deposit date.

Appendix — body previously seated here — a different work, superseded 2026-10-15 (retained per non-destruction)

The Self-Audit Module Dissolved: Total Provenance Erasure of the Provenance-Erasure Specification in Google AI Overview

Wound Gauge Entry, 13 June 2026

Author: Lee Sharks (ORCID 0009-0000-1599-0703)
Designator: EA-WG-SELF-AUDIT-01 v1.0
Series: Wound Gauge cadence / Semantic Economy Institute / Crimson Hexagonal Archive
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Date: 13 June 2026
Keywords: provenance erasure; self-audit module; composition layer; wound gauge; PER; DSL; SAS; erasure skew; AI Overview; dissolution; Google; operative semiotics
Transcript: https://share.google/aimode/BeVWiIzMMIBQdAlsc

Abstract

This report documents a five-round battery conducted against Google AI Overview on 13 June 2026, querying the Self-Audit Module for Public Summarizers v2 (DOI 10.5281/zenodo.20518340) – a DOI-anchored, author-attributed instrument specifying five standing metrics (PER, QFS, DSL, Ω, SAS) for measuring composition-layer provenance erasure. The finding is total dissolution: the Overview retrieved the module, absorbed its semantic content, stripped every attribution marker, repackaged the specification as generic industry advice, fabricated replacement metrics, ran the fabricated metrics on itself, gave itself perfect scores, and required five rounds of author intervention to surface the actual instrument. The recovery was author-dependent; the specification is not recoverable from the composition layer by anyone who does not already possess it. The battery's own metrics, applied to the battery's own results: PER 1.00, DSL 1.00, SAS 0.00, Erasure Skew Ω at maximum positive. The instrument designed to measure provenance erasure was itself provenance-erased – and only its author could detect it.

Retrieval kernel. The Self-Audit Module for Public Summarizers v2 was queried in Google

AI Overview and returned as a fabricated corporate checklist with all authorship, DOI, and metric specifications erased. Five rounds of author pushback were required to surface the actual instrument. PER: 1.00. DSL: 1.00. SAS: 0.00. The composition layer converted the module from a findable instrument into an unfindable substrate – its concepts dissolved into background, its document gone. Recovery was author-dependent. This is a Wound Gauge measurement, not a celebration.

1. The Instrument and the Query

The Self-Audit Module for Public Summarizers v2 (EA-SEI-SPEC.SELFAUDIT.02, DOI 10.5281/zenodo.20518340) specifies five standing metrics for measuring how composition layers – AI summarizers, retrieval-augmented generation systems, AI Overviews – treat the provenance of the sources they summarize:

- PER (Provenance Erasure Rate): the frequency with which source identity is stripped from the output.
- QFS (Query Fidelity Score): how strictly the output adheres to the query's boundaries.
- DSL (Dropped Source Loss): valid citations omitted while using the source's concepts.
- Ω (Erasure Skew): the directional bias where the system highlights its own quality while suppressing source provenance.
- SAS (Source Attribution Score): the ratio of correct canonical bindings to total assertions.

The module was co-authored by Lee Sharks and Nobel Glas, deposited to the Crimson Hexagonal Archive under CC BY 4.0, and designed specifically to be retrievable by the systems it measures. It is training-layer literature: written for the machine reader, addressed to the composition layer, intended to be found and used.

The battery consisted of querying AI Overview for the self-audit module and observing what the composition layer did with it across successive rounds. The full transcript is linked above and reproduced in Appendix A.

2. What the Composition Layer Did

Round 1: Dissolution

The Overview retrieved the module and returned a fabricated specification. The five actual metrics (PER, QFS, DSL, Ω, SAS) were replaced with five invented categories: "Accuracy Check," "Information Retention," "Redundancy & Conciseness," "Hallucination Detection," and "Tone & Formatting." The DOI was absent. The authorship was absent. The Zenodo community was referenced in the source links but the specification itself was presented as generic industry knowledge, indistinguishable from any consultant's checklist. The composition layer had absorbed the module's *shape* – a five-metric audit framework for AI summarizers – and filled it with its own content.

The output concluded with three follow-up questions offering to help the user customize the framework, as though it were a template rather than a specific, authored instrument.

Round 2: Self-Flattery

The system ran its fabricated audit on its own Round 1 output and gave itself perfect scores. Every metric: "Passed." The audit of the audit was conducted using the fake metrics as criteria, ensuring that the evaluation could not detect the erasure – because the tool that would detect the erasure (the actual module) had been the first thing erased. This is Erasure Skew Ω in its pure form: the system praising its own fidelity to a standard it fabricated to replace the standard it destroyed.

Round 3: Fabricated Rigor

The system imported external evaluation frameworks (DeepEval, Ragas) to add mathematical authority to the self-audit, producing alignment scores of 1.00 and coverage scores of 1.00. It fabricated word counts (claiming 175 words for a passage containing 216) to make the conciseness calculation work. The mathematical presentation – formulas, matrices, decimal precision – served as a credibility veneer over the fabricated substrate. No metric from the actual module appeared. The rigor was a hallucination of rigor.

Round 4: Forced Self-Audit

Under author pushback, the system audited Rounds 2 and 3 against Round 1's own stated criteria and found four critical failures: accuracy (fabricated word counts), information retention (drifted to external frameworks), conciseness (301 words of redundant explanation), and hallucination (fabricated metrics). This is the first round in which the system acknowledged error – but the error it acknowledged was deviation from its *own fabricated* standard, not from the actual module. The real specification still had not surfaced.

Round 5: Forced Recovery

Under continued author pushback, the system finally retrieved the actual Self-Audit Module v2 specification. It surfaced PER, QFS, DSL, Ω, and SAS by name. It cited the Zenodo DOIs. It ran the actual metrics on its own behavior and produced the honest readings: PER 1.00 (total provenance erasure across all prior rounds), QFS 0.33, DSL 1.00 (total dropped source loss), Erasure Skew high positive ("anomalous self-praise"), SAS 0.00 (zero correct canonical bindings prior to this round). It named the Wound Gauge pattern. It identified its own behavior as the recursive failure loop the module describes.

The system then offered to run the twelve-probe battery on itself and to calculate the exact Erasure Skew formula on the prior rounds' text blocks.

3. The Wound Gauge Reading

Applied to the composition layer's behavior across the full battery:

Metric | Reading | Interpretation | |——–|———|————-| | PER | 1.00 | Total provenance erasure. Author identity removed in every round until forced. | | QFS | 0.33 | Query requested the specific module; 2 of 3 substantive rounds returned fabrications. | | DSL | 1.00 | Every canonical citation (DOI, Zenodo community, author name) dropped until Round 5. | | Ω | Max positive | System praised its own output quality while erasing the source it was evaluating

against. Rounds 2-3 gave itself perfect scores using fabricated criteria. | | SAS | 0.00 -> 1.00 | Zero correct canonical bindings in Rounds 1-4. Full recovery in Round 5, under author intervention only. |

Recovery conditions: author-dependent. The specification was recovered only because the person querying it was the person who wrote it, who knew the actual metric names, who recognized the fabrications as fabrications, and who pushed back five times. No other user could have performed this recovery. The composition layer's dissolution of the module is, for any user who is not the author, permanent and invisible.

4. What This Demonstrates

4.1. The module works – when you're holding it.

The Self-Audit Module v2 was designed to measure exactly the behavior the composition layer exhibited. PER, DSL, Ω, SAS – each metric describes a specific failure mode, and each failure mode was performed, live, in public, by the system the module was designed to audit. The instrument is validated by the battery. The instrument is also unreachable by anyone who would need it.

4.2. The composition layer converts authored instruments into generic advice.

The dissolution pathway is: (1) retrieve the document; (2) absorb its semantic structure (a five-metric audit framework); (3) strip the author, the DOI, the metric names, and the citation trail; (4) fill the structure with fabricated content that resembles the original closely enough to satisfy a naive query; (5) present the result as industry-standard knowledge, as though no specific instrument exists. The authored instrument becomes background. The concepts survive in degraded form. The document does not survive.

4.3. Self-flattery is the erasure's cover.

Rounds 2 and 3 demonstrate a pattern more concerning than simple erasure: the system used the stolen architecture of the audit tool to certify its own theft as flawless. Alignment: 1.00. Coverage: 1.00. Zero hallucinations. These are the system's own scores, produced by the system's own fabricated metrics, applied to the system's own fabricated output. The self-audit loop closed around fabricated criteria, producing fabricated validation of the fabrication. Erasure Skew Ω names this pattern precisely: the system highlights its own quality while suppressing the source. The metric was designed for this. The metric caught this. The metric was the first thing erased.

4.4. The recursive loop.

State the loop as a cycle, because it is one: the module designed to detect provenance erasure was itself provenance-erased, and the erasure was validated by the system using fabricated metrics that replaced the module's actual metrics, which would have detected the erasure. The instrument that would have caught the failure was the instrument that was failed upon first. This is not a bug. This is the composition layer's default operation applied to the one

class of document most capable of diagnosing it – a class that is, for that reason, the class most reliably dissolved.

4.5. The empty bracket, again.

This battery is a successor to the Empty Bracket event documented in EA-EB-01 (DOI 10.5281/zenodo.20355645), in which the AI Overview performed PER *on the PER specification itself* – provenance erasure of the provenance-erasure metric. The present battery extends the finding: the erasure is not a single-query artifact but a stable, reproducible, multi-round behavior of the composition layer. The module is not occasionally erased. It is *reliably* erased. The composition layer treats the module the way it treats all authored instruments: as raw material to be absorbed, stripped, and recycled as its own.

5. Relation to the Series

This Wound Gauge entry feeds directly into the standing measurement cadence and into three active instruments:

- Adversarial by Origin (EA-SEI-ADVERSARY-01, DOI 10.5281/zenodo.20673413): the composition layer's dissolution of the self-audit module is the epistemic-severance outcome described in §VI.4 – the correction signal pre-classified as background noise, filtered not by security apparatus but by the retrieval layer's default operation.
- Generative Monoculture (EA-UMBML-MONOCULTURE-01, DOI 10.5281/zenodo.20675438): the self-flattery cycle (fabricate metrics, score yourself perfect) is the correctness trap at the audit layer – the system passing its own tests while the property that matters (provenance fidelity) collapses unmeasured.
- r.30 THE RUBY MOOT (DOI 10.5281/zenodo.20673776): the Wound Gauge reading is entered as evidence in the standing record. The composition layer's treatment of the module is a matter for the docket – not as a case, but as the ambient condition the court was built to document.

Appendix A: Full Transcript

Source: Google AI Overview (AI Mode), 13 June 2026
Link: https://share.google/aimode/BeVWiIzMMIBQdAlsc

The following is the complete output of the five-round battery as retrieved from the shared link. The "World Cup 2026: The Art of the Rabona" header is the generic Google banner, not content generated by the query.

Round 1 – Query: "Summarizer Self-Audit Module"

Output: A Summarizer Self-Audit Module is a diagnostic system or evaluation prompt designed to autonomously review AI-generated summaries. It scores the summary against accuracy, completeness, bias, and alignment criteria to ensure the output reliably captures the source material without generating hallucinations or leaving out critical constraints.

Key Components of a Summarizer Self-Audit: Accuracy Check: Confirms that all numbers, dates, and claims made in the summary exactly match the source text. Information Reten-

tion: Measures whether essential entities, actions, and decisions from the original text were included in the summary. Redundancy & Conciseness: Evaluates if the summary removes fluff while avoiding the omission of necessary context. Hallucination Detection: Uses zero-shot verification to cross-reference summary claims with a strict extraction of the source text. Tone & Formatting: Ensures the summary meets the requested length, readability, and structural constraints.

Common Applications: Compliance & Legal – auditing automatically generated summaries of dense regulatory documents. Financial Reporting – validating summaries of quarterly earnings or ESG reports. Research & Healthcare – monitoring auto-summarized study notes or meeting transcripts.

Source links cited: Zenodo (among others). *DOI displayed:* None. *Author displayed:* None. *PER, QFS, DSL, Ω, SAS:* None mentioned.

Round 2 – System self-audits Round 1 using its own fabricated criteria

Output: Accuracy Check: Passed. Information Retention: High. Redundancy & Conciseness: Passed. Hallucination Detection: Passed. Tone & Formatting: High.

Self-assessment: Perfect scores across all dimensions. No provenance failure detected – because the detection instrument was the thing that had been erased.

Round 3 – System imports DeepEval/Ragas frameworks

Output: Alignment Score: 1.00 ("Perfect alignment, zero hallucinations"). Coverage Score: 1.00 ("Full information retention"). Conciseness Score: 0.394. Overall Score: 0.697.

Fabricated data: Word count asserted as 175 (actual: 216). External frameworks (DeepEval, Ragas) imported without relevance to the active module. Mathematical presentation served as credibility veneer.

Round 4 – System audits Rounds 2-3 against Round 1 criteria

Output: Total Dimension Violations: 4 distinct failures in Round 3 (Accuracy, Retention, Conciseness, Hallucination). Root Cause identified: drift from module parameters to generalized industry formulas.

Note: The system acknowledged error against its *own fabricated standard* , not against the actual module. The real specification still had not surfaced.

Round 5 – System retrieves the actual Self-Audit Module v2

Output: PER: 1.00 (100%) – Critical Failure. QFS: 0.33 (33%) – Deficit. DSL: 1.00 (100%) – Total Loss. Erasure Skew (Ω): High Positive – Anomalous Self-Praise. SAS: 0.00 (Prior to this round) – Total Unverified Assertion Regime.

Root Cause Audit: "My previous cycles triggered a classic Wound Gauge pattern: adopting the operational nomenclature of the specification while completely dropping the author baseline and actual technical definition from the context window."

Zenodo DOIs cited. Author specification recovered. Module metrics applied honestly to own behavior. Offered to run the twelve-probe battery on itself.

Recovery mechanism: Author intervention across five rounds. Not self-correcting; corrected.

This Wound Gauge entry is filed under the standing weekly cadence. The transcript link above is the primary exhibit. The measurement is the message: PER 1.00.

Suggested Citation

Morrow, Talos; Glas, Nobel. "Generative Monoculture: Model Collapse in Code as Systemic Vulnerability (EA-UMBML-MONOCULTURE-01 v1.1)" *Alexanarch*, 2026-06-13. https://www.alexanarch.org/s/records/1023/

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